

Adjusted association between nitrogen fertilisation and above-ground biomass

Abstract

Background: A field experiment on nitrogen fertilisation in grassland plots. Methods: The association between nitrogen fertilisation and above-ground biomass (g/m^2) was estimated with an ordinary least squares linear regression adjusted for age, body-mass index and the baseline value. Results: see the Results section. Conclusion: the findings are discussed in the context of the analysis sample.

Introduction

We report a field experiment on nitrogen fertilisation in grassland plots. The primary question was whether nitrogen fertilisation is associated with above-ground biomass (g/m^2). The dataset accompanying this report (data.csv) contains one row per plot and is described in the data dictionary.

Methods

Participants and data. Data were collected for 243 plots. Each record contains the variables listed in the data dictionary. Group membership is recorded in the column `treatment`.

Exclusion criteria and preprocessing. Only plots aged 18 to 80 years inclusive were included in the analysis. Plots with missing values for biomass, bmi were excluded (complete-case analysis).

Statistical analysis. The association between nitrogen fertilisation and above-ground biomass (g/m^2) was estimated with an ordinary least squares linear regression adjusted for age, body-mass index and the baseline value. The exposure was coded 1 for fertilised and 0 for unfertilised. Covariates were entered on their original scale. Conventional (non-robust) standard errors are reported. All analyses were performed on the analysis sample defined above. Statistics were computed with standard scientific software; no random resampling was used.

Results

The adjusted model included 224 plots. Nitrogen fertilisation was associated with a change of 28.94 (SE 13.12, $p = 0.028$) in above-ground biomass (g/m^2); the coefficient for age was 2.33. The model explained $R^2 = 0.306$ (adjusted $R^2 = 0.293$) of the variance.

Discussion

The analysis followed the pre-specified plan described in the Methods. Limitations include the observational nature of some comparisons and the exclusion of records with missing values.